Vignesh N Salian

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[Vignesh-Salian]

PROFILE SUMMARY

Information Science Engineering student with strong proficiency in Python and experience in web development using Flask. Skilled in Machine Learning and Computer Vision using libraries such as Scikit-learn, NumPy, Pandas, and OpenCV. Experienced in developing practical applications including real-time vision-based systems and web-based tools. Passionate about applying AI and software development skills to solve real-world problems.

EDUCATION

NMAM Institute of Technology, Aug 2023 – May 2027 | Karnataka, India
B. Tech. in Information Science and Engineering; GPA: 8.47

SKILLS SUMMARY

Programming Languages: Python, JavaScript

Web Development: HTML, CSS, Flask

Machine Learning: Scikit-learn, NumPy, Pandas

Computer Vision: OpenCV

Databases: MySQL, MongoDB

Tools & Technologies: Git, GitHub, VS Code

Soft Skills: Analytical Thinking, Attention to Detail, Problem Solving, Team Collaboration

PROJECTS

ArUco Marker-Based Distance Measurement System Oct – 2025
- Built a real-time Python and OpenCV system for nasal–jaw distance measurement.
- Implemented camera calibration and ArUco markers for millimeter-level accuracy.
- Achieved ~150 FPS with only 4.9% CPU usage.

VidSnapAI: AI Short-Form Video Generator Feb–2026
- Built a Flask web application to generate short-form vertical videos from uploaded media.
- Integrated ElevenLabs TTS and FFmpeg for automated voiceover and video processing.
- Designed an asynchronous background pipeline for scalable media generation.

ML Benchmark: Credit Card Fraud Detection Mar–2026
- Built a machine learning pipeline for credit card fraud detection.
- Implemented SMOTE and evaluated multiple classification models.
- Developed a Streamlit dashboard for model benchmarking and fraud prediction.

PUBLICATIONS

Computer Vision System for Real-Time Nasal–Jaw Distance Measurement in Dental Applications
- Published in IEEE ICCES 2025, PSG Institute of Technology, Coimbatore, India (Indexed in IEEE Xplore).
- Presents a computer vision system using ArUco markers and OpenCV for real-time nasal–jaw distance measurement.